Maximum Principles for Linear Second Order Elliptic Operators

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In these notes we have collected results on maximum principles for linear second order elliptic operators, in some cases modifying proofs for the Laplacian to simplify the exposition. We reproduce proofs of various maximum principles for classical solutions, strong solutions, and viscosity solutions. We will also demonstrate a few applications of these maximum principles. The primary references are Gilbarg and Trudinger [6] and Evans [4].

We are considering operators of the form:

$$a_{ij}\partial_{ij} + b_i\partial_i + c \tag{1}$$

where ∂_i indicate partial derivatives, and repeated indices are summed over. Ellipticity means that the coefficient matrix a_{ij} is positive definite, and uniform ellipticity means that the ratio of the maximum to minimum eigenvalue of a_{ij} is bounded. Setting a_{ij} to the identity gives us the Laplace operator Δ. Throughout the notes Ω will indicated a bounded open subset of $\mathbb{R}^n$. We will assume some familiarity with elliptic equations, and may miss a few definitions along the way.

Note: The original purpose of these notes was to understand the tools used to prove maximum principles for different types of solutions to elliptic equations, with the hope that we might be able to use a discrete analogue of one of these tools to study high order accurate finite difference methods.

Chapter 1

Classical Solutions

Classical solutions satisfy the PDE with derivatives in the elliptic operator defined in the usual sense. We will first discuss a maximum principle (MP) which applies to the Laplace operator. Proofs in this section are from [6] and [4].

1.1 Mean-value MP

A function u is called **harmonic** in Ω if at each point in that domain it satisfies $\Delta u = 0$. u is called **subharmonic** or **superharmonic** if we replace that equality with $\geq$ or $\leq$, respectively. We will only worry about subharmonic functions because they can be negated to obtain superharmonic functions.

Harmonic functions satisfy a mean value property, meaning the value of u at any point x in Ω is equal to the average of u over any ball centered at x which is contained in Ω, as well as the average over the boundary of any such ball. We will prove this fact using a change of variables:

THEOREM 1.1.1. *Suppose $u \in C^2(\Omega)$ is harmonic in Ω. Then for any ball $B_R(x) \subset\subset \Omega$, we have*

$$u(x) = \frac{1}{|B_R(x)|} \int_{B_R(x)} u(y)dy \tag{1.1}$$

$$u(x) = \frac{1}{|\partial B_R(x)|} \int_{\partial B_R(x)} u(y)dy \tag{1.2}$$

Proof: Let $\epsilon \in (0, R)$. By the divergence theorem, we have:

$$0 = \int_{B_\epsilon(x)} \Delta u(y)dy = \int_{\partial B_\epsilon(x)} \nabla u(y) \cdot n \; dS(y) \tag{1.3}$$

where n is the outward facing normal and dS is the surface measure on the ball. Let $r := |y - x|$, and $\theta = \frac{y-x}{r}$. Note $\frac{dS}{d\theta} = \epsilon^{n-1}$. We can rewrite the right hand side above as:

$$0 = \int_{\partial B_\epsilon(x)} \nabla u(y) \cdot n \; dS(y) = \int_{\partial B_\epsilon(x)} \partial_r u(x + \epsilon\theta(y)) dS(y) \tag{1.4}$$

$$= \int_{|\theta|=1} \partial_r u(x + \epsilon\theta) d\theta \; \epsilon^{n-1} \tag{1.5}$$

$$= \epsilon^{n-1} \partial_\epsilon \int_{|\theta|=1} u(x + \epsilon\theta) d\theta \tag{1.6}$$

$$= \epsilon^{n-1} \partial_\epsilon \left[\epsilon^{1-n} \int_{\partial B_\epsilon(x)} u(y) dS(y) \right] \tag{1.7}$$

meaning the expression on the final line in brackets is constant as $\epsilon \to R$, or

$$\epsilon^{1-n} \int_{\partial B_\epsilon(x)} u(y) dS(y) = R^{1-n} \int_{\partial B_R(x)} u(y) dS(y) \tag{1.8}$$

Now taking $\epsilon \to 0$, we have

$$\lim_{\epsilon \to 0} \epsilon^{1-n} \int_{\partial B_\epsilon(x)} u(y) dS(y) = \lim_{\epsilon \to 0} \epsilon^{1-n} |\partial B_\epsilon(x)| \left(\frac{1}{|\partial B_\epsilon(x)|} \int_{\partial B_\epsilon(x)} u(y) dS(y) \right) = \omega_n n u(y) \tag{1.9}$$

where ω_n is the unit ball volume and $n\omega_n$ is the unit ball surface volume. Putting these together, we have:

$$u(y) = \frac{1}{n\omega_n R^{n-1}} \int_{\partial B_R(x)} u(y) dS(y) = \frac{1}{|\partial B_R(x)|} \int_{\partial B_R(x)} u(y) dy \tag{1.10}$$

To get the equality over the solid ball, integrate the surface equality from 0 to R. □

Analogous results can be obtained for subharmonic functions by replacing the equalities in Theorem 1.1.1 with $\leq$, and likewise for superharmonic functions. We can remember the terminology *sub*-harmonic because u is *below* its local average.

One can see how this property encodes a MP: u cannot achieve a local maximum because it must be equal to its average locally.

THEOREM 1.1.2. *Suppose $u \in C^2(\Omega)$ is harmonic in Ω, and that there exists a point x in Ω where $u(x) = \sup_\Omega u$. Then u is constant in Ω.*

Proof: Define Ω' as the (nonempty) subset of Ω on which u achieves its supremum, which we set to be M. By the continuity of u, Ω' is relatively closed in Ω. At any point y in Ω', by the mean value property we must have that, for any $B_R(y) \subset\subset \Omega$, $u = M \in B_R(y)$. Thus Ω' is relatively open in Ω (recall this means that Ω' is the intersection of an open set and Ω). Therefore $\Omega' = \Omega$, and $u = M$ in Ω. □

Note that we can use the same technique to prove that a harmonic function which achieves its infimum in Ω must be constant. The above MP is called a *strong* MP because it implies a *weak* MP:

THEOREM 1.1.3. *Suppose $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ is harmonic in Ω. Then for any $x \in \Omega$*

$$\inf_{\partial\Omega} u \leq u(x) \leq \sup_{\partial\Omega} u \tag{1.11}$$

Proof: By 1.1.2, we have that if u achieves its supremum in Ω, then it must be constant, which by continuity implies $\sup_\Omega u = \sup_{\partial\Omega} u$. If u does not achieve its supremum in Ω, then it must achieve its supremum on $\partial\Omega$. We can apply the same argument to obtain the lower bound. □

Using similar arguments, we can show that subharmonic functions satisfy a strong MP, meaning if a subharmonic function achieves its supremum in Ω, it must be constant. Subharmonic functions also satisfy the weak MP:

$$\Delta u \geq 0 \in \Omega \implies \sup_{\Omega} u = \sup_{\partial\Omega} u \tag{1.12}$$

From these MPs, along with a Green's function for the ball, one can build an entire uniqueness and existence theory for solving the Dirichlet problem. For details see Chapter 2 of [6]. An interesting fact proved in this chapter is that the mean value property is in fact equivalent to harmonicity. Thus we must use other tools to prove MPs for more general elliptic operators, and in doing so will find another method for proving the MP for the Laplace operator.

1.2 Calculus-Based MPs

If L is an elliptic operator, we will call a function u which satisfies $Lu \geq 0$ in a domain Ω a **subsolution**. We will let $\nu_0(x), \nu_n(x)$ indicate the smallest and largest eigenvalues, respectively, of the coefficient matrix a_{ij}.

THEOREM 1.2.1. *Let L be an elliptic operator with continuous coefficients and $c = 0$, such that the ratio $|b_i|/\nu_0$ is uniformly bounded above by b_0. Assume $u \in C^2(\Omega) \cap C(\overline{\Omega})$. Then if $Lu \geq 0$, $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$.*

Proof: Suppose $Lu > 0 \in \Omega$ and u achieves its maximum at the interior point x_0. Then at that point $\nabla u(x_0) = 0$ and $D^2u(x_0)$ is negative semidefinite. We can locally map the second order term to a Laplacian by an orthogonal change of coordinates and a rescaling, so at x_0 it suffices to just consider $L = \Delta$ (the gradient term will be zero). Then we have

$$\Delta u(x_0) > 0 \quad \implies \quad diag(D^2u(x_0)) > 0 \tag{1.13}$$

which is a contradiction. Now relax the assumption to $Lu \geq 0$ and consider $u^\epsilon(x) = u + \epsilon e^{\lambda x_1}$ for some $\lambda > 0$ and $\epsilon > 0$. We can see for a large enough λ that

$$Le^{\lambda x_1} = \left(a_{11}\lambda^2 + \lambda b_1\right) e^{\lambda x_1} \geq \nu_0 \left(\lambda^2 - \lambda b_0\right) > 0 \tag{1.14}$$

For all $\epsilon > 0$ we can use the first part of the proof to show that $\max_{\overline{\Omega}} u^\epsilon = \max_{\partial\Omega} u^\epsilon$, so we drive $\epsilon \to 0$ to achieve the desired conclusion. When c is nonpositive, we can show that $Lu \geq 0 \in \Omega \implies \sup_\Omega u \leq \sup_{\partial\Omega} u^+$. □

An interesting result that can be derived from this principle is Hopf's lemma:

THEOREM 1.2.2. *Let L be a uniformly elliptic operator with continuous coefficients and $c = 0$. Let $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$, and suppose $Lu \geq 0$. Additionally suppose there exists an $x_0 \in \partial\Omega : u(x_0) > u(x)\ \forall\ x \in \Omega$. Lastly, suppose there exists a ball $B \in \Omega$ such that $x_0 \in \partial B$. Then $\frac{\partial u}{\partial \nu}(x_0) > 0$.*

This result is interesting because it says at the maximum boundary point the gradient flows strictly outward. Flipping signs, this means that a positive supersolution which is zero on the boundary will have strictly inward gradient flow on the entire boundary; this means that $u > 0$ strictly. To get the basic idea of the proof, we consider the Laplace operator; for the general case see [4].

Proof:

Assume $c \geq 0$, and let r be the radius of B and shift so B is centered at 0. Now let $v = e^{-\lambda|x|^2} - e^{-\lambda r^2}$ for some λ that we are going to choose (this would come in to play to dominate the b term). We compute:

$$-\Delta v = -\lambda^2 e^{-|x|^2}|x|^2 \leq 0 \tag{1.15}$$

and it is strictly negative in $R := B(0,r) \setminus B(0,r/2)$. Now by assumption there exists an ϵ with

$$u(x_0) \geq u(x) + \epsilon v(x) \ \in \partial B(0,r/2) \tag{1.16}$$

and clearly $u(x_0) \geq u(x) + \epsilon v(x)$ on $\partial B(0,r)$ because v is $0 \in \partial B(0,r)$. We can say

$$-\Delta(u + \epsilon v - u(x_0)) \leq 0 \ \in \ R \tag{1.17}$$

since $u - \epsilon v - u(x_0) \leq 0$ on ∂R, we can see that $u + \epsilon v - u(x_0) \leq 0 \in R$ by the weak MP. This implies that, since $u(x_0) - \epsilon v(x_0) - u(x_0)$ is zero on ∂B, we must have $\frac{\partial u(x_0)}{\partial \nu} + \epsilon \frac{\partial v(x_0)}{\partial \nu} \geq 0$, or

$$\frac{\partial u(x_0)}{\partial \nu} \geq -\epsilon \frac{\partial v(x_0)}{\partial \nu} = 2\epsilon r \lambda e^{-\lambda r^2} > 0 \tag{1.18}$$

which proves the claim. □

The Strong MP follows from Hopf's lemma:

THEOREM 1.2.3. *Let Ω be connected. Then if $u \in C^2(\Omega) \cap C(\overline{\Omega})$ and $Lu \geq 0$ and u achieves its maximum over $\overline{\Omega}$ in the interior of Ω, u must be constant.*

Proof: Assume that we have such a u which is nonconstant; denote Ω' as the subset of Ω on which u is less than its maximum value, which by assumption is nonempty. Then there must exist a ball $B(x_0) \in \Omega'$ for which some point y in ∂B lies in the complement of Ω', meaning $u(y)$ is a maximum. By Hopf's lemma, $\nabla u(y) \neq 0$ at that point, which is only possible if y also lies in $\partial\Omega$, meaning u cannot achieve an interior maximum. □

In Chapter 3 of [6], the authors utilize these MPs to prove various apriori estimates for solutions of elliptic equations. It is in fact possible to obtain regularity estimates using MPs; a few examples are given in [6] and the more general case is given in [2].

1.3 Spectral Analysis

In this section we review several results on spectral properties of elliptic operators that do not use Hilbert space theory, which is of particular interest for the analysis of finite difference schemes.

NOTE: In this section we follow the convention in [4], in which an elliptic operator L has a negative sign on the second order term. By nonsymmetric operator we mean in the sense of the L^2 inner product.

From [4] 6.5.2:

THEOREM 1.3.1. *Let L be a possibly nonsymmetric elliptic operator with $c \geq 0$, and let Ω be a domain with smooth boundary, and let all coefficients be smooth. Then I) there exists a real eigenvalue $\lambda_1 > 0$ whose real part is less than or equal to the real part of all other eigenvalues. II) The corresponding eigenfunction is positive within Ω and III) the eigenvalue λ_1 is simple.*

Proof of I:

Define the Banach space X to be $H^m(\Omega) \cap H^1_0(\Omega)$, where $m = [n/2] + 3$, which by some combination of Morrey's inequality and Sobolev inequalities, we have $X \subset C^2(\overline{\Omega})$. Define $A : X \to X := L^{-1}$. Now define the cone $C := \{u \in X : u \geq 0 \in \Omega\}$. By the weak MP, A maps C to C ($Lu \geq 0 \implies u \geq 0$).

Pick any $w \in C$, and let define $v := Aw$, or $Lv = w$. Find a number μ such that $\mu v \geq w$. Now choose some $\epsilon > 0$ and consider the equation

$$u = \eta A\left[u + \epsilon w\right], \text{ or } Lu = \eta\left[u + \epsilon w\right] \tag{1.19}$$

where we consider solutions $u \in C$.

Our goal will be to drive ϵ to 0 to obtain an eigenfunction. We will now show that if (1.19) has a solution, then $\eta \leq \mu$:

Suppose we have found a $u \in C$ with $u = \eta A[u + \epsilon w]$. This means that we have

$$u \geq \eta A \epsilon w = \eta \epsilon v \tag{1.20}$$
$$\geq \frac{\eta \epsilon}{\mu} w \tag{1.21}$$

where we used the fact that $u \geq 0$. We can also use the positivity of w to say:

$$u \geq \eta A u \geq \frac{\eta^2 \epsilon}{\mu} A w \tag{1.22}$$
$$= \frac{\eta^2 \epsilon}{\mu} v \geq \frac{\eta^2}{\mu^2} \epsilon w \implies \tag{1.23}$$
$$u \geq \eta A u \geq \frac{\eta^3 \epsilon}{\mu^2} A w = \frac{\eta^3 \epsilon}{\mu^2} v \geq \frac{\eta^3}{\mu^3} \epsilon w \implies \tag{1.24}$$
$$u \geq \frac{\eta^k}{\mu^k} \epsilon w \; \forall \; k \in \mathbb{N} \tag{1.25}$$

which, since u, w are bounded, implies that $\mu \geq \eta$. Note that we again used the MP (and linearity) to say $Ax \geq Ay$ when $x > y \geq 0$.

Our next task is to show there exist some solutions to (1.19) for certain η. Define $S_\epsilon = \{u \in C : \; \exists \; 0 \leq \eta \leq 2\mu : u = \eta A[u + \epsilon w]\}$. We next claim that S_ϵ is unbounded (meaning it has arbitrarily large elements) in X:

Suppose that S_ϵ is bounded, meaning there exists some b_ϵ with $\|u\|_X < b_\epsilon$ for all $u \in S_\epsilon$. Fix η. Since the operator $u \to 2\eta A[u + \epsilon w]$ is continuous and compact (comes from embeddings and energy estimates) from $C \to C$, and the set C is a convex subset of X, then by a version of Schaefer's fixed point theorem [4] 9.2.2, there exists a fixed point $u = \eta A[u + \epsilon w]$, which contradicts our earlier claim.

Now since S_ϵ is unbounded, we can drive ϵ to zero and make a sequence $\eta_\epsilon, v_\epsilon$ with $\|v_\epsilon\|_X \geq 1/\epsilon$, and $v_\epsilon = \eta_\epsilon A[v_\epsilon + \epsilon w]$ and let u_ϵ be $v_\epsilon / \|v_\epsilon\|_X$. Since A is a compact mapping, there exists a convergent subsequence with $\eta_{\epsilon_k} \to \eta$, and $u_{\epsilon_k} \to u$. In the limit $\epsilon_k \to 0$, we have $u = \eta A u$, which we will rewrite as $Lw_1 = \lambda_1 w_1$, and w_1 is positive by the MP.

This gives us a different characterization of the principal eigenvalue than the Hilbert space approach (min-max theorem for Rayleigh quotients) - it comes as the result of a convergent subsequence of a sequence which drives solutions to our strange equation (1.19) to be very large.

We now must show that this principal eigenvalue bounds the real parts of all of the other eigenvalues below. Suppose we have a complex eigenpair λ, u. Choose a positive, smooth real-valued function w and let $v = u/w$. I think this proof can be understood by considering $L = -\Delta$; we can note below where ellipticity would be used in the more general case.

Define the operator K, which satisfies

$$Kv + \left(\frac{Lw}{w} - \lambda\right) v = 0 \implies \tag{1.26}$$

$$Kv := -\Delta v - \frac{2}{w}\nabla w \cdot \nabla v \tag{1.27}$$

We then have

$$K(|v|^2) = K(vv^*) = -v^*\Delta v - 2\nabla v \cdot \nabla v^* - v\Delta v^* - \frac{2}{w}\nabla w \cdot \nabla(vv^*) \tag{1.28}$$

$$\leq v^*Kv + vKv^* \tag{1.29}$$

$$= |v|^2 \left(\lambda - \frac{Lw}{w} + \lambda^* - \frac{Lw}{w}\right) = 2|v|^2 \left(Re(\lambda) - \frac{Lw}{w}\right) \tag{1.30}$$

note, in the general case ellipticity would have given this same bound because the gradient term from the chain rule would be positive definite.

Now consider $w := w_1^{1-\epsilon}, 0 < \epsilon < 1$. We have

$$Lw_1^{1-\epsilon} = \frac{(1-\epsilon)}{w_1^{\epsilon}} Lw_1 + \frac{\epsilon(1-\epsilon)}{w_1^{1+\epsilon}} \left(\nabla w_1 \cdot \nabla w_1\right) \tag{1.31}$$

$$\geq (1-\epsilon)\lambda_1 w \tag{1.32}$$

which gives us

$$\frac{Lw}{w} \geq (1-\epsilon)\lambda_1 \tag{1.33}$$

which implies

$$K(|v|^2) \leq 2|v|^2 \left(Re(\lambda) - (1-\epsilon)\lambda_1\right) \tag{1.34}$$

Now if $Re(\lambda) - (1-\epsilon)\lambda_1 \leq 0$, then $K(|v|^2) \leq 0$, which implies $v = 0$ by the MP, which holds for K. Thus $Re(\lambda) \geq (1-\epsilon)\lambda_1 \geq 0$ for all $\epsilon > 0$, meaning $Re(\lambda) \geq \lambda_1$. $\square$

We note here that we obtained some strictly positive terms by the chain and product rule, NOT by integration by parts. This proof uses a couple aspects of ellipticity: the positivity of the principal eigenfunction, the MP, and the definition of ellipticity, which produces a nonnegative gradient norm term.

Proof of II: see I.

Proof of III: (λ_1 is simple)

Assume there exists some v with $Lv = \lambda_1 v$; WLOG we can assume its real and positive in some part of Ω. Let $\chi := \sup\{\eta > 0 : w_1 - \eta v \geq 0 \in \Omega\} < \infty$. Then let $u = w_1 - \chi v$, which is nonnegative, and we have

$$Lu = \lambda_1 u \geq 0 \in \Omega \ \ , \ u = 0 \in \partial\Omega \tag{1.35}$$

by the Strong MP and Hopf's lemma we have $u > 0$, $\frac{\partial u}{\partial \nu} < 0 \in \partial\Omega$, meaning we can find a small ϵ with $u \geq \epsilon v$, meaning $w_1 - (\chi + \epsilon)v \geq 0$, which contradicts the definition of χ. Therefore λ_1 is simple. □

We now look at another characterization of the principal eigenvalue of an elliptic operator L with $c \geq 0$:

THEOREM 1.3.2. *Let λ_1 be the principal eigenvalue of L. Then define $X := C^\infty(\overline{\Omega}) \cap \{u > 0 \in \Omega, u = 0 \in \partial\Omega\}$. Then we have:*

$$\lambda_1 = \sup_{u \in X} \inf_{x \in \Omega} \frac{Lu(x)}{u(x)} \tag{1.36}$$

Proof: Clearly $\lambda_1 \leq \sup_{u \in X} \inf_{x\in\Omega} \frac{Lu(x)}{u(x)}$. Define $f(u) := \inf_{x\in\Omega} \frac{Lu(x)}{u(x)}$. Then $f(u)u \leq Lu$. Let w_1^* be the principal eigenfunction of L^*, which we showed was positive. We can compute:

$$f(u) \int_\Omega w_1^* u \leq \int_\Omega w_1^* Lu \tag{1.37}$$

$$= \lambda_1 \int_\Omega u w_1^* \tag{1.38}$$

which implies $f(u) \leq \lambda_1$ over X, or $\sup_{u\in X} \inf_{x\in\Omega} \frac{Lu(x)}{u(x)} \leq \lambda_1$. □

This is the analogue of the Collatz-Wielandt formula. For a nonnegative c we can just add a constant so that the shifted operator has a positive principal eigenfunction, and we can see that the constant will get subtracted from both sides.

We are applying a tool from functional analysis called the Krein-Rutman theorem [7], which is a generalization of the Perron-Frobenius theorem. Let X be a Banach space. A "cone" K is a closed convex subset of X that is closed under multiplication by nonnegative scalars. (ex: nonegative functions in $W^{k,p}$) A *total cone* means that $K - K$ is dense in X. The Krein-Rutman theorem says:

THEOREM 1.3.3. *Let $K \subset X$ be a closed total cone. Let T be a compact linear operator from $X \to X$ that satisfies $T(K) \subset K$, and has positive spectral radius $\rho(T)$. Then T has a positive eigenvector $v \in K \setminus 0$ with $Tv = \rho(T)v$.*

Chapter 2

Strong Solutions

This material is Ch. 9 of [6]. Strong solutions live in $W^{2,p}$, meaning derivatives in the operator are defined in the generalized sense. In this section we will prove yet another maximum principle that holds for classical solutions, and by a proper limiting process show that the results also hold for strong solutions.

2.1 Aleksandroff Maximum Princple

Define the upper contact set Γ^+ of u:

$$\Gamma^+ = \{y \in \Omega : \exists p : u(x) \leq u(y) + p \cdot (x - y) \ \forall \ x \in \Omega\} \tag{2.1}$$

If $u \in C^1(\Omega)$, then p will always be Du. Define the normal mapping $\chi_u(y) : \Omega \to \mathbb{R}^n$:

$$\chi_u(y) = \{p \in \mathbb{R}^n : u(x) \leq u(y) + p \cdot (x - y) \ \forall \ x \in \Omega\} \tag{2.2}$$

For example, if we let u be a cone with height a at the origin and base of radius R. Then $\chi_u(0) = B_{a/R}(0)$. First we prove a preliminary version of the max principle:

THEOREM 2.1.1. *Let $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$, and $d = diam(\Omega)$. Then*

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u + \frac{d}{\omega_n^{1/n}} \left(\int_{\Gamma^+} |\det D^2 u| \right)^{1/n} \tag{2.3}$$

Proof: WLOG assume $u \leq 0 \in \partial\Omega$. We have

$$\begin{aligned} |\chi_u(\Omega)| &= |\chi_u(\Gamma^+)| && (2.4) \\ &= \int_{y \in Du(\Gamma^+)} dy && (2.5) \\ &\leq \int_{\Gamma^+} |\det D^2 u| && (2.6) \end{aligned}$$

which follows from the "area formula," a generalization of the change of variables formula. Suppose that u takes its maximum value of M at some point $y \in \Omega$, and let w be a cone with vertex of height M and base of radius d. We claim that $\chi_w(\Omega) \subset \chi_u(\Omega)$: consider the function $L(x) = m + v \cdot (x - y)$, where $|v| < m/d$. Then we have that $u(y) = L(y)$, and $L > u$ on $\partial\Omega$, meaning that $\max_{\overline{\Omega}} u - L$ occurs inside Ω, so $D(u - v) = 0$ at that point. Geometrically, for each upper supporting hyperplane of w, there is a parallel supporting hyperplane to u. Therefore we have that

$$\omega_n \frac{M^n}{d^n} = |\chi_w(\Omega)| \leq |\chi_u(\Omega)| \leq \int_{\Gamma^+} |\det D^2 u| \implies \tag{2.7}$$

$$\sup_{\Omega} u = M \leq \frac{d}{\omega_n^{1/n}} \left(\int_{\Gamma^+} |\det D^2 u| \right)^{1/n} \tag{2.8}$$

We now use a matrix inequality $\det X \det Y \leq \left(\frac{Tr(XY)}{n}\right)^n$ to obtain, on the set Γ^+, on which $D^2 u \leq 0$, that

$$|\det D^2 u| = \det(-D^2 u) \leq \frac{1}{\det A} \left(\frac{-a^{ij} D_{ij} u}{n} \right)^n \tag{2.9}$$

where $A = a_{ij}$. Thus using the previous theorem we have

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u + \frac{d}{n\omega_n^{1/n}} \left\| \frac{a^{ij} D_{ij} u}{(\det A)^{1/n}} \right\|_{L^n(\Gamma^+)} \tag{2.10}$$

Ultimately we would like to prove a maximum principle for a nondivergence form ellipitc operator with nonpositive c, so we need a way to deal with the other terms. More generally, we can use the area formula to obtain

$$\int_{Du(\Gamma^+) = \chi_u(\Omega)} g \leq \int_{\Gamma^+} g(Du) |det D^2 u| \tag{2.11}$$

where g is nonegative and locally integrable. Using the same argument as above, we have

$$\int_{B_M(0)} g \leq \int_{\Gamma^+} g(Du) |\det D^2 u| \leq \int_{\Gamma^+} g(Du) \left(\frac{-a^{ij} D_{ij} u}{n} \right)^n \tag{2.12}$$

which proves the claim. □

Now assume we have an elliptic operator of the form $Lu = a^{ij}(x) D_{ij} u + b_i D_i u + cu$ where $c \leq 0$, ν_0, ν_n are the min and max eigenvalues of A, $\mathcal{D}^* := (\det A)^{1/n}$, and $\frac{|b|}{\mathcal{D}^*}, \frac{|f|}{\mathcal{D}^*} \in L^n(\Omega)$. Our goal is to prove:

THEOREM 2.1.2. *Let L be as above, with $Lu \geq f$ for $u \in C^0(\Omega) \cap W^{2,n}_{loc}(\Omega)$. Then we have*

$$\sup_{\Omega} u \leq \sup_{\partial\Omega} u^+ + C \left\| \frac{f}{\mathcal{D}^*} \right\|_{L^n(\Omega)} \tag{2.13}$$

where C depends on the dimension, Ω, and $\frac{|b|}{\mathcal{D}^*}$.

If we look at the formula (2.12), we want to choose a g so that the integral on the left comes out to something that we can isolate M from, and the integral on the right looks like the norm of f times some constant related to b. Let $\mu > 0$ and set $g(p) = (|p|^{n/(n-1)} + \mu^{n/(n-1)})^{1-n}$. On the set where u is positive, since $c \le 0$, we have that if $Lu \ge f$ implies

$$\left(\frac{-a^{ij}D_{ij}u}{n\mathcal{D}^*}\right) \le \frac{b^iD_iu - f}{n\mathcal{D}^*} \tag{2.14}$$

$$\le \frac{|b||Du| + |f|}{n\mathcal{D}^*}\frac{g(Du)^{1/n}}{g(Du)^{1/n}} \tag{2.15}$$

$$\le \frac{(|b|^n + \mu^{-n}|f|^n)^{1/n}}{ng(Du)^{1/n}\mathcal{D}^*} \tag{2.16}$$

Applying (2.12) to this we have

$$\int_{B_M(0)} g \le \frac{1}{n^n}\int_{\Gamma^+}(|b|^n + \mu^{-n}|f|^n)/\mathcal{D} \tag{2.17}$$

For the integral on the left we can use the fact that

$$g(p) = (|p|^{n/(n-1)} + \mu^{n/(n-1)})^{1-n} \ge 2^{2-n}(|p|^n + \mu^n)^{-1} \tag{2.18}$$

to obtain

$$\int_{B_M(0)} g \ge \omega_n \log\left(\frac{M^n}{\mu^n} + 1\right)2^{2-n} \implies \tag{2.19}$$

$$\omega_n \log\left(\frac{M^n}{\mu^n} + 1\right)2^{2-n} \le \frac{1}{n^n}\int_{\Gamma^+}(|b|^n + \mu^{-n}|f|^n)/\mathcal{D} \implies \tag{2.20}$$

$$M \le \mu\left(\exp\left[\frac{2^{n-2}}{n^n\omega_n}\int_{\Gamma^+}(|b|^n + \mu^{-n}|f|^n)/\mathcal{D}\right] - 1\right)^{1/n} \tag{2.21}$$

Now choosing $\mu = \|f/\mathcal{D}^*\|_{L^n(\Gamma^+)}$ we have

$$M \le \left(\exp\left[\frac{2^{n-2}}{n^n\omega_n}\int_{\Gamma^+}\left(1 + \frac{|b|^n}{\mathcal{D}}\right) - 1\right]\right)^{1/n}\|f/\mathcal{D}^*\|_{L^n(\Gamma^+)} \tag{2.22}$$

let C be the constant before the norm. In order to extend this to the case that $u \in W^{2,n}_{loc}(\Omega)$, let the sequence $\{u_m\}$ converge in $W^{2,n}_{loc}$ to u. If we choose an $\epsilon > 0$, let $\Omega_\epsilon \subset\subset \Omega$ be a domain on which $u_m \in \partial\Omega_\epsilon \le \epsilon + \sup_{\partial\Omega} u$. By the maximum principle above we have

$$\sup_{\Omega_\epsilon} u_m \le \epsilon + \sup_{\partial\Omega} u^+ + \frac{C}{\nu_0}\left\|a^{ij}D_{ij}(u_m - u) + b^iD_i(u_m - u)\right\|_{L^n(\Omega_\epsilon)} + C\left\|f/\mathcal{D}^*\right\|_{L^n(\Omega_\epsilon)} \tag{2.23}$$

from which we recover the desired result from the convergence of u_m, which is uniform on the compact sets Ω_ϵ. $\square$

Taking a step back, this result says that for a quite general operator L, we can bound the max value of u in terms of boundary values, the source term and the coefficients. The key to this proof in my mind is that we relate the maximum value of u to its second derivatives by 1) considering the contact set and 2) considering the gradient as a mapping of space. Note if $f = 0$ we just get the typical calculus-based maximum principle.

Chapter 3

Viscosity Solutions

These results are from [3].

Let us define "viscosity" solutions of the PDE $F(x, u(x), Du, D^2u) =: F[u] = 0$, which must make sense for actual C^2 solutions. A function u will be a subsolution (think $\Delta u \geq 0$) in the viscosity sense if it is upper semicontinuous and if there exists a ball $B_r(x) \subset \Omega$ and a C^2 function φ such that $\varphi(x) = u(x)$ and $\varphi \geq u \in B_r(x)$, then $F(x, \varphi(x), D\varphi(x), D^2\varphi(x)) \geq 0$. Think of φ as test functions which we are transferring derivatives to. Switch inequalities to get a supersolution (must be LSC); a viscosity solution is both a subsolution and super solution.

How is this notion consistent with an actual solution? Suppose we have $u \in C^2, F[u] \geq 0$. If we have a x, φ that fits our definition, we will have $D\varphi(x) = Du(x)$ and $D^2\varphi(x) \geq D^2u(x)$ (this means that $D^2\varphi(x) - D^2u(x)$ is positive semidefinite, which we get from the fact we have reached a local maximum of $u - \varphi$. We have the structural assumption that $F(A, ...) \geq F(B, ...)$ when $A \geq B$, so we get $F(x, \varphi(x), D\varphi(x), D^2\varphi(x)) \geq F(x, u(x), Du(x), D^2u(x))$. If we have a C^2 viscosity solution, then we can just use it directly as a test function.

A different formulation of viscosity solutions is used in [3]. Define the *jet* $J^{2,+}_{\mathcal{O}}u(x)$, where u is USC, as the set of $(p, X) \in \mathbb{R} \times \mathcal{S}(N)$ such that $u(y - x) \leq u(x) + p \cdot (y - x) + \frac{1}{2}(y - x)^T X(y - x) + o(\|y - x\|^2)$. A viscosity subsolution satisfies $F(x, u(x), p, X) \geq 0$ for all elements of the jet.

3.1 Comparison Principle

In this section we review the proof of the main analytical result of this theory, which allows one to prove a comparison principle for semicontinuous sub and super solutions. This ties everything together; we have a weak definition of what it means to be a solution, so a concern is that this might be too weak to guarantee uniqueness. The proof is a bit difficult to understand so I am going to write it out here.

THEOREM 3.1.1. *Let $\mathcal{O}_i$ be a locally compact subset of $\mathbb{R}^{N_i}$ for $i = 1...k$, and $\mathcal{O} = \prod_i \mathcal{O}_i$. Let $u_i \in USC(\mathcal{O}_i)$, and φ be twice continuously differentiable in a neighborhood of $\mathcal{O}$. Set*

$w(x) = \sum_i u_i(x_i)$, and suppose $\hat{x}$ is a local maximum of $w - \varphi$ over $\mathcal{O}$. Then for $\epsilon > 0$, there exists $X_i \in \mathcal{S}(N_i)$ such that $(D_{x_i}\varphi(\hat{x}_i), X_i) \in \overline{J}^{2,+}_{\mathcal{O}} u_i(\hat{x}_i)$ for $i = 1...k$, and we have the following inequality:

$$-\left(\frac{1}{\epsilon} + \|A\|\right) I \leq \begin{bmatrix} X_1 & ... & 0 \\ ... & & \\ 0 & ... & X_k \end{bmatrix} \leq (A + \epsilon A^2) \tag{3.1}$$

where $A = D^2\varphi(\hat{x})$.

Notes: These inequalities refer to the ordering of symmetric matrices. The jet closure is defined as: $(p, X) \in \overline{J}^{2,+}_{\mathcal{O}} u(x)$ if there exists a sequence $(x_n, u(x_n), p_n, X_n) \to (x, u(x), p, X)$ with $(p_n, X_n) \in J^{2,+}_{\mathcal{O}} u(x_n)$. We define the jet $J^{2,+}_{\mathcal{O}} u(x)$ as the set of $(p, X) \in \mathbb{R} \times \mathcal{S}(N)$ such that $u(y - x) \leq u(x) + p \cdot (y - x) + \frac{1}{2}(y - x)^T X (y - x) + o(\|y - x\|^2)$

Big picture: we want to relate elements of jets of a collection of functions to each other using an auxillary smooth function. This is a perturbation of the result that would hold if the w was smooth.

Proof: We allow the USC functions to take on the value $-\infty$ to simplify the problem statement. We first assume $\mathcal{O}_i = \mathbb{R}^{N_i}$ by setting u_i to $-\infty$ off of a compact neighborhood of $\hat{x}_i$. Next we can translate coordinates to put the maximum at the origin, and we can also assume $\varphi = \frac{1}{2}x^T A x$, aka it is a pure quadratic: Replace $\varphi(x)$ by $\varphi(x)\varphi(0) - D\varphi(0) \cdot x$, and $u_i(x)$ by $u_i(x) - u_i(0) - D_{x_i}\varphi(0) \cdot x$. Then setting $A = D^2\varphi(0)$, we can see that $\varphi(x) = \frac{1}{2}x^T A x + o(\|x\|^2)$ and we know that $w(x) - \varphi(x) \leq 0$. We can add a small diagonal to A to get $w(x) - \frac{1}{2}x^T(A + \eta I)x < 0$ near the origin. If we again localize, then we will that $w(x) - \frac{1}{2}x^T(A + \eta I)x$ has a strict local maximum at the origin. If we have proved the theorem in this case for each $\eta > 0$, then we can take the limit to get the full result, and later we can just add back in the slope $D\varphi(0)$ to obtain an element of the jet closure.

Firstly, we would like to regularize the USC function in way so that we can obtain something with second derivatives; then we can obtain information via a Hessian test. However, this process means that our regularized function may not sit underneath the same quadratic function. Let us introduce a new coordinate ξ; this is where the regularized version of w is going to live. The following matrix inequality can be proved using CS:

$$x^T A x \leq \xi^T (A + \epsilon A^2)\xi + \left(\frac{1}{\epsilon} + \|A\|\right) |x - \xi|^2 \tag{3.2}$$

This is interesting because it says $x^T A x$ is underneath a shifted normal quadratic centered at ξ, and that same normal quadratic, as a function of ξ, when flipped over and shifted is underneath another quadratic at the origin. This can be utilized to give the bound

$$w(x) - (\lambda/2)|x - \xi|^2 \leq (1/2)\xi^T (A + \epsilon A^2)\xi \tag{3.3}$$

when $\lambda = \frac{1}{\epsilon} + \|A\|$. Now define

$$\hat{w}(\xi) = \sup_{x \in \mathbb{R}^N} \left(w(x) - \frac{\lambda}{2}|x - \xi|^2\right) \tag{3.4}$$

and likewise define $\hat{u}_i(\xi_i)$. The functions $\hat{w}$ and $\hat{u}_i$ are semiconvex, which means that $\hat{w}(\xi) + \frac{\lambda}{2}|\xi|^2$ is convex:

$$\hat{w}(\xi) + \frac{\lambda}{2}|\xi|^2 = \sup_{x\in\mathbb{R}^N}\left(w(x) - \frac{\lambda}{2}|x-\xi|^2\right) + \frac{\lambda}{2}|\xi|^2 = \sup_{x\in\mathbb{R}^N}\left(w(x) - \frac{\lambda}{2}|x|^2 + \lambda x^T\xi\right) \quad (3.5)$$

so this object is the supremum of a family of planes, meaning it is convex. We can see from our definition of the sup convolution that if this function is second differentiable anywhere, we will have $-\lambda I \leq D^2\hat{w}$ by convexity.

Stopping here for a second: we have a semiconvex regularization of $\hat{w}$ which we have shown sits underneath a perturbed version of our original quadratic x^TAx. Our hope is that we can find points close to the origin where our regularized function is second differentiable; if so we will have control over these second derivatives. We would then like to 1) relate these derivatives to the jets of the u_i and 2) drive the points toward the origin to create a sequence that satisfies our jet closure definition.

To do this we use two facts about semiconvex functions: Firstly, Aleksandrov's theorem says they are a.e. second differentiable. Secondly, we have Jensen's lemma: Let φ be semiconvex with a strict local maximum at $\hat{x}$. Then for $r, \delta > 0$, the set

$$K = \{y \in B_r(\hat{x}) : \exists p \in B_\delta(0) : \varphi(x) + p\cdot x \text{ has local maximum at } y\} \quad (3.6)$$

has positive measure. This lemma is in the spirit of the Aleksandrov maximum principle. The key idea is $B_\delta \subset D\varphi(K)$, because at a local maximum of $\varphi(x) + p\cdot x$, $D\varphi = -p$, and semiconvexity gives a uniform upper bound on $|detD^2\varphi(x)|$ I will skip these proofs for now. Using these two facts about semiconvex functions, we can prove this lemma:

Lemma: Let $f \in C(\mathbb{R}^N)$, B symmetric, and f λ semiconvex and $\max_{\mathbb{R}^N} f(\xi) - \frac{1}{2}\xi^TB\xi = f(0)$; then there exists X symmetric such that $(0, X) \in \overline{J}^2 f(0)$, and $\lambda I \leq X \leq B$. By this closure we mean have actual second derivatives forming the sequence.

Proof: Make the maximum strict by subtracting $|\xi|^4$, so we can apply Jensen's lemma, meaning for all $\delta > 0$ we have a $q_\delta, |q_\delta| < \delta$ such that $f(\xi) - \frac{1}{2}\xi^TB\xi - |\xi|^4 + \xi^Tq_\delta$ has a local maximum in $B_\delta(0)$ at the point ξ_δ AND importantly, by Aleksandrov's theorem, at this point we will be second differentiable. This means we have the bounds

$$-\lambda I \leq D^2f(\xi_\delta) \leq B + O(\delta^2) \quad (3.7)$$

with $(Df(\xi_\delta), D^2f(\xi_\delta)) \in J^2f(\xi_\delta)$, and now we take a subsequential limit to get the result. We see that $Df(\xi_\delta) = O(\delta) \to 0$. □

We now want to apply this lemma to our sup-conv $\hat{w}$. By doing so we obtain $(0, X) \in \overline{J}^2\hat{w}(0)$, after setting $B = (A + \epsilon A^2)$, then split the X diagonally to obtain the bound stated in the theorem: we have $(0, X_i) \in \overline{J}^2\hat{u}_i(0)$. The last step is to show that we can transfer an element of this jet to a jet of u_i. We will see that this can be done because the definition of the sup-conv means that we "pull" jets from the function we are regularizing, and as we increase λ these points get very close.

Lemma: If $(q, Y) \in J^{2,+}\hat{v}(\eta)$, then $(q, Y) \in J^{2,+}v(\eta + q\ \lambda)$, and $\hat{v}(\eta) + (1/2\lambda)|q|^2 = v(\eta + q/\lambda)$.

Proof: Let y be the point that $\hat{v}(\eta)$ comes from. Then we have

$$v(x) - \frac{\lambda}{2}|x - \xi|^2 \leq \hat{v}(\xi) \tag{3.8}$$

$$\leq v(y) - \frac{\lambda}{2}|y - \eta|^2 + q \cdot (\xi - \eta) + \frac{1}{2}(\xi - \eta)^T Y(\xi - \eta) + o(|\xi - \eta|^2) \tag{3.9}$$

$$= v(y) - \frac{\lambda}{2}|y - \eta|^2 + q \cdot (\xi - \eta) + O(|\xi - \eta|^2) \tag{3.10}$$

Now setting $\xi = x - y + \eta$ we get

$$v(x) \leq v(y) + q \cdot (x - y) + \frac{1}{2}(x - y)^T Y(x - y) + o(|x - y|^2) \tag{3.11}$$

Meaning we have q, Y in the jet at y. To solve for y, set $x = y$ and $\xi = \eta + \alpha(\lambda(\eta - y) + q)$. Then we get

$$0 \leq \alpha|\lambda(\eta - y) + q|^2 + O(\alpha^2) \tag{3.12}$$

Driving $\alpha < 0$ small we get $y = \eta + q/\lambda$.

Finally, if we have $(q_n, Y_n) \in J^{2,+}\hat{v}(\xi_n)$, then $(q_n, Y_n) \in J^{2,+}v(\xi_n + q_n/\lambda)$. If we assume that $(\xi_n, \hat{v}(\xi_n), q_n, Y_n) \to (0, 0, 0, Y)$, we only need to make sure $\lim_n v(\xi_n + q_n/\lambda) \to v(0)$. By upper semicontinuity, we have

$$v(0) \geq \limsup_n v(\xi_n + q_n/\lambda) \limsup_n \left(\hat{v}(\xi_n) + \frac{1}{2\lambda}|q_n|^2\right) = \hat{v}(0) \geq v(0) \tag{3.13}$$

Therefore we have $(0, Y) \in \overline{J}^{2,+}v(0)$. We are not increasing the regularization here: for a fixed λ, if we happen to have a jet closure with a zero slope element, then our actual USC function has this same jet closure. The ϵ at the beginning of the proof, as well as the norm of A, control the amount of regularization we need to get the bound stated in the theorem. □

Now, we will try to walk through this proof backwards in order to understand why we did different things. If things are smooth, we can prove a similar statement (without the lower bound) by directly using a Hessian test at $\hat{x}$. The regularization serves as a tool to get a function which is second differentiable a.e., and has the other important property (Jensen's lemma) that near a maximum point, there is a set of positive measure on which a small perturbation of the regularized function achieves its maximum near the original maximum point. We can make these perturbations arbitrarily small; this means we can more or less find something that behaves like the regularized function but has two derivatives there. The "magic" of this regularization is that if something is in the jet of the regularized function, it must have come from the jet of the original function. One of the key tricks used in many different ways throughout this proof is to compare two functions by putting them each on their own space; we can freeze one coordinate at a time at traverse the other function to compare that function to a single point of the other one, or compare them both to something else.

Now we want to see exactly how we can interpret this theorem as a comparison principle. Recall that a subsolution u satisfies, for each $p, X \in \overline{J}_\Omega^{2,+}u(x)$, $F(x, u(x), p, X) \geq 0$.

THEOREM 3.1.2. *Let F be elliptic and proper, and let u be USC, v LSC be a viscosity subsolution and supersolution, respectively. Then if $u \leq v \in \partial\Omega, u \leq v \in \Omega$.*

Proof: Define

$$M_\alpha = \sup_{\Omega\times\Omega} \left(u - v - \frac{\alpha}{2}|x-y|^2\right) \tag{3.14}$$

As we increase α, this should give us the maximum of $u-v$ - we are penalizing any movement off of the diagonal $x = y$. Now suppose $M_\alpha < \infty$, and there exist (x_α, y_α) such that

$$\lim_{\alpha\to\infty} \left(M_\alpha - \left(u(x_\alpha) - v(y_\alpha) - \frac{\alpha}{2}|x_\alpha - y_\alpha|^2)\right)\right) = 0 \tag{3.15}$$

Then it can be shown that this M gives us what we want, i.e. if $\hat{x}$ is a limit point of x_α

$$\lim_{\alpha\to\infty} \alpha|x_\alpha - y_\alpha|^2 = 0 \tag{3.16}$$

$$\lim_{\alpha\to\infty} M_\alpha = u(\hat{x}) - v(\hat{x}) = \sup_{\Omega\times\Omega} u - v \tag{3.17}$$

We satisfy these assumptions since we are working with semicontinuous functions on a compact domain, so x_α, y_α will be where the maximum in the definition of M_α is achieved. Let us assume that there is a point z at which $u(z) - v(z) = \delta > 0$. Then we have $M_\alpha \geq \delta$, and for large enough α, we will have $x_\alpha, y_\alpha \in \Omega$, because $u \leq v \in \partial\Omega$. Our goal is to show, using the theorem on sums, that $M_\alpha \geq \delta$ contradicts our assumptions about F.

Now we will apply the Theorem on Sums: let $k = 2, u_1 = u, u_2 = -v$, $\mathcal{O}_1 = \mathcal{O}_2 = \Omega$, and let $\varphi = \frac{\alpha}{2}|x-y|^2$. Then we have

$$D_x\varphi = \alpha(x-y) = -D_y\varphi \tag{3.18}$$

$$D^2\varphi = \alpha \begin{bmatrix} I & -I \\ -I & I \end{bmatrix} \tag{3.19}$$

which by application of the theorem, we have $(\alpha(x-y), X) \in \overline{J}_\Omega^{2,+}u(x_\alpha)$ and $(\alpha(x-y), Y) \in \overline{J}_\Omega^{2,-}v(y_\alpha)$, and the inequality

$$-\left(\frac{1}{\epsilon} + 2\alpha\right)\begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \leq \begin{bmatrix} X & 0 \\ 0 & -Y \end{bmatrix} \leq (\alpha + 2\epsilon\alpha^2)\begin{bmatrix} I & -I \\ -I & I \end{bmatrix} \tag{3.20}$$

holds for all $\epsilon > 0$. A nice choice is $\epsilon = 1/\alpha$, which gives

$$-3\alpha \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \leq \begin{bmatrix} X & 0 \\ 0 & -Y \end{bmatrix} \leq 3\alpha \begin{bmatrix} I & -I \\ -I & I \end{bmatrix} \tag{3.21}$$

recall that this is an inequality in terms of eigenvalues, or equivalently in terms of inner products. The right hand side turns constant vectors into zero, from which we can obtain $X \leq Y$. Because these matrices are in the jet closures of the sub and supersolution u, v, we should have that

$$F(x_\alpha, u(x_\alpha), \alpha(x_\alpha - y_\alpha), X) \leq 0 \tag{3.22}$$
$$F(y_\alpha, v(y_\alpha), \alpha(x_\alpha - y_\alpha), Y) \geq 0 \tag{3.23}$$

Now let us assume the F has a particular form of properness: there exists a positive γ such that

$$\gamma(r - s) \leq F(x, r, p, X) - F(x, s, p, X) \text{ when } s \leq r \tag{3.24}$$

For example, we could have $F(x, r, p, X) = -Tr(X) + cr$ for a positive c (this is the Laplace operator plus a zeroth order term). Now we can say

$$\gamma\delta \leq \gamma(u(x_\alpha) - v(y_\alpha)) \tag{3.25}$$
$$\leq F(x_\alpha, u(x_\alpha), \alpha(x_\alpha - y_\alpha), X) - F(x_\alpha, v(x_\alpha), \alpha(x_\alpha - y_\alpha), X) \tag{3.26}$$
$$\leq [F(x_\alpha, u(x_\alpha), \alpha(x_\alpha - y_\alpha), X) - F(y_\alpha, v(y_\alpha), \alpha(x_\alpha - y_\alpha), Y)] \tag{3.27}$$
$$+ [F(y_\alpha, v(y_\alpha), \alpha(x_\alpha - y_\alpha), Y) - F(x_\alpha, v(x_\alpha), \alpha(x_\alpha - y_\alpha), X)] \tag{3.28}$$
$$\leq F(y_\alpha, v(y_\alpha), \alpha(x_\alpha - y_\alpha), Y) - F(x_\alpha, v(x_\alpha), \alpha(x_\alpha - y_\alpha), X) \tag{3.29}$$

now we must make an additional structural assumption that when (3.21) holds,

$$F(y, r, \alpha(x - y), Y) - F(x, r, \alpha(x - y), X) \leq \omega(\alpha|x - y|^2 + |x - y|) \tag{3.30}$$

where $\omega : [0, \infty] \to [0, \infty]$ approaches zero from the right. Since we showed that $\lim_{\alpha\to\infty} \alpha|x_\alpha - y_\alpha|^2 = 0$, we will reach the contradiction $\gamma\delta \leq 0$. □

This additional assumption looks a bit weird but in fact it is stronger than ellipticity; see [3] for details.

Chapter 4

Finite Difference Solutions

We can utilize a discrete form of the maximum principle to study finite difference solutions to the Poisson equation. We will need some new notation:

Let $h > 0$ be the grid spacing parameter. Points $i \in \mathbb{Z}^n$ with components i_d will be the indices of points $x_i := ih \in \mathbb{R}^n$ with components $x_{i,d}$. We let B_r denote the closed ball on $\mathbb{Z}^n$ of radius r, or $\{i \in \mathbb{Z}^n : |i|_{\ell^2} \leq r\}$, and ∂B_r denote the set of all lattice points on the boundary of ∂B_r. ∂B_r will be empty unless $r^2 = \sum_{d=1}^n i_d^2, i \in \mathbb{Z}^n$. Let Q_r be the closed cube on $\mathbb{Z}^n$ of radius r, or $\{i \in \mathbb{Z}^n : |i|_{\ell^\infty} \leq r\}$. $B_r(i), Q_r(i)$ will indicate translations of these sets. If Ω is a bounded, connected open set in $\mathbb{R}^n$, we let Ω^h indicate the set of all grid points x_i which are contained inside Ω. When not specified, assume $h = 1$.

We will work with various notions of a discrete boundary $\partial\Omega^h$ throughout this work. For this section, we will consider a sharp representation of the boundary as used in immersed boundary type methods, and let it indicate the set of points in $\mathbb{R}^n$ where Ω intersects the "gridlines" of the lattice, or lines of the form $x_d = i \in \mathbb{Z}$ for $n \geq 2$, and in one dimension $\partial\Omega^h$ is simply the points which define the interval.

Let $\ell^\infty(\Omega^h)$ be the space of bounded real valued functions on Ω^h. Let u_i denote the value of the grid function u^h at the grid point x_i. Let T_k be the translation operator:

$$T_k u_i = u_{i+k} \tag{4.1}$$

and the discrete derivative operators:

$$\partial^{1,h} := \frac{1}{2h}\left(T_1 - T_{-1}\right) \tag{4.2}$$

$$\partial^{2,h} := \frac{1}{h^2}\left(T_{-1} - 2T_0 + T_1\right) \tag{4.3}$$

Let $\Delta^h : \ell^\infty(\Omega^h) \to \ell^\infty(\Omega^h)$ denote the discrete Laplacian operator:

$$\Delta^h := \sum_{d=1}^{n} \partial_d^{2,h} \tag{4.4}$$

where $\partial_d^{2,h}$ indicates a discrete partial derivative in the d coordinate direction. This stencil has an important property:

A discrete operator is **positive type** if the center coefficient is negative and all other coefficients are nonnegative.

We consider x_i to be an "interior" point if the indices $i \pm e_d$ are all contained within Ω^h. The definition of an interior point will change throughout this work, but generally means a point whose stencil does not involve any boundary information. Near the boundary, there are several different approaches for constructing finite difference stencils. We typically must make some geometric assumptions on $\partial\Omega$ relative to the grid spacing. For now we assume the grid spacing is fine enough and the boundary curvature is such that there is no more than one intersection of $\partial\Omega$ and a lattice gridline between adjacent lattice points. In addition, we assume that if $i \in \Omega^h$, then for all d, at least one of $i + e_d, i - e_d$ is also in Ω^h.

Now consider an index i which is not an interior point, meaning at that point the boundary $\partial\Omega$ intersects at least one of the line segments between i and one of its $2n$ nearest neighbors. We assume that u^h is known at all points of $\partial\Omega^h$. The simplest approaches for constructing a finite difference stencil for ∂_d^2 at these points are one dimensional, in that we construct the finite difference stencils for each coordinate direction separately. We give two examples for such procedures, which both can be understood in one dimension. The first procedure is used, for example, in [1].

Firstly, we can consider the boundary value along with $k \geq 2$ interior values and fit a degree k polynomial to these points. The finite difference operator at x_i will be the the second derivative of this polynomial, which will be some linear combination of these $k + 1$ values.

Secondly, we can extend the discrete function u^h to points outside the domain such that we can simply apply the interior stencil to the extended function; these exterior points are typically called "ghost" or "fictitious" points. Consider the grid point i on a one dimensional lattice. Suppose the left boundary lies between $i - 1$ and i, and let $0 < \kappa \leq 1$ denote the distance between the boundary and i. We can fit a degree k polynomial through the boundary value at $x_i - h\kappa$ and k interior points; for example, an order $k = 0$ extension would simply set $u_{i-1} = u(x_i - h\kappa)$, and an order $k = 1$ extension would set $u_{i-1} = u(x_i - h\kappa) + \frac{u(x_i - h\kappa) - u_i}{\kappa}(1 - \kappa)$. We then recover a stencil near the boundary by applying Δ^h to the extended function in each coordinate direction to obtain a stencil which only involves boundary and interior values. It is important to note that here that this extension is used for the purposes of building a one dimensional finite difference approximation to the second derivative at x_i, so it may be the case that this extended value is defined differently in different coordinate directions. Setting $k = 2$ is equivalent to using $k = 2$ interior points for the extension procedure; however in general there is a great deal of flexibility in how we perform this polynomial interpolation procedure for higher order accurate methods, and it can be useful to think of it as conceptually distinct from the extension methods.

For this second order case, we will consider only the $k = 1$ and $k = 2$ extensions. Again consider the situation in which the boundary lies between $i - 1$ and i. We have the one dimensional stencils:

$$k = 1 : \frac{1}{h^2}\left(\frac{1}{\kappa}u(x_i - h\kappa) - \frac{1+\kappa}{\kappa}u_i + u_{i+1}\right) \tag{4.5}$$

$$k = 2 : \frac{1}{h^2}\left(\frac{2}{\kappa^2+\kappa}u(x_i - h\kappa) - \frac{2}{\kappa}u_i + \frac{2}{\kappa+1}u_{i+1}\right) \tag{4.6}$$

Note that these stencils are both of positive type. They also have the property that they are positive on quadratic functions. For $k = 2$, the stencil is exact on quadratic functions, and for $k = 1$, we have that:

$$\frac{1}{h^2}\left(\frac{1}{\kappa}(x_i - h\kappa)^2 - \frac{1+\kappa}{\kappa}x_i^2 + (x_i + h)^2\right) = 1 + \kappa^2 \tag{4.7}$$

Using either a $k = 1$ or $k = 2$ one dimensional extension procedure, we can prove a discrete maximum principle which is often first credited to [5]:

THEOREM 4.0.1. *Suppose Δ^h is positive type at each point in Ω^h. Then if $\Delta^h u^h \geq 0 \in \Omega^h$, $\max_{\overline{\Omega^h}} u^h = \max_{\partial\Omega^h} u^h$*

Proof: First suppose that $\Delta^h u^h > 0 \in \Omega^h$. Since Δ^h is positive type at each index i in Ω^h, we have that $\Delta^h u_i^h > 0 \implies u_i < u_j$ for some $j \in \overline{\Omega^h}$. Therefore it must be the case that $\max_{\overline{\Omega^h}} u^h = \max_{\partial\Omega^h} u^h$.

Now suppose $\Delta^h u^h \geq 0 \in \Omega^h$. Let v^h be the restriction of the function x_1^2 to $\overline{\Omega^h}$. Then we have for any $\epsilon > 0$ that $\Delta^h(u + \epsilon v) > 0 \in \Omega^h$ which implies $\max_{\overline{\Omega^h}} u^h + \epsilon v^h = \max_{\partial\Omega^h} u^h + \epsilon v^h$, and can send $\epsilon \to 0$ to obtain the desired result. □

We can use this to prove a stability result.

THEOREM 4.0.2. *Let $u^h \in \ell^\infty(\overline{\Omega^h})$. Then there exists a constant C which depends only on diam(Ω) such that*

$$|u^h|_{\ell^\infty(\Omega^h)} \leq C|\Delta^h u^h|_{\ell^\infty(\Omega^h)} + |u^h|_{\ell^\infty(\partial\Omega^h)} \tag{4.8}$$

Proof: Let $M := |\Delta^h u^h|_{\ell^\infty(\Omega^h)}$ and $N := |u^h|_{\ell^\infty(\partial\Omega^h)}$. Suppose that Ω lies in the slab $0 < x_1 < D$. Let v^h be the restriction of the function

$$-Mx_1^2 + MD^2 + N \tag{4.9}$$

If we are using a $k = 2$ extension procedure, we will have

$$\Delta^h v^h = -2M \in \Omega^h \tag{4.10}$$

and if we are using a $k = 1$ extension we will have at non interior points which are a fraction κ from the boundary in the x direction:

$$\Delta^h v^h = -M(1 + \kappa^2) \leq -M \tag{4.11}$$

Therefore we have

$$\Delta^h(u^h - v^h) \geq -M + M \geq 0 \tag{4.12}$$

and

$$u^h - v^h \leq N + MD^2 - MD^2 - N \leq 0 \in \partial\Omega^h \tag{4.13}$$

Applying Theorem 4.0.1, we have

$$\max_{\Omega^h}(u^h - v^h) \leq \max_{\partial\Omega^h} u^h - v^h = 0 \implies \tag{4.14}$$

$$u^h \leq v^h \leq -Mx_1^2 + MD^2 + N \leq MD^2 + N \tag{4.15}$$

Now applying the same result to $u^h + v^h$, we have

$$u^h \geq -N - MD^2 \tag{4.16}$$

which gives us

$$|u^h|_{\ell^\infty(\Omega^h)} \leq D^2|\Delta^h u^h|_{\ell^\infty(\Omega^h)} + |u^h|_{\ell^\infty(\partial\Omega^h)} \tag{4.17}$$

$\square$

We now show that this maximum principle can be used to prove that the discrete solution converges in a certain sense to a classical solution of the exact PDE.

Let $u \in C^k(\Omega) \cap C^{k-1}(\overline{\Omega})$ where $k \geq 5$, and $\partial\Omega$ satisfies our geometric assumptions. At interior points x_i, we have by Taylor's Theorem:

$$u(x) = u(x_i) + \sum_{|p| \leq k-1} \frac{D^p u(x_i)}{p!}(x - x_i)^p + C \left\|(x - x_i)^k\right\|_{\ell^\infty} \tag{4.18}$$

$$\Delta^h u(x_i) = \Delta u(x_i) + \frac{h^2}{12}\sum_{d=1}^{n} \partial_d^4 u(x_i) + O(h^2) \tag{4.19}$$

where p is a multi-index. For a point x_i near the boundary, let κ_d be the distance fraction to the boundary in direction d, which could possibly be 1. We have for the $k = 1$ extension procedure

$$\Delta^h u(x_i) = \sum_{d=1}^{n} \frac{\kappa_d + 1}{2}\partial_d^2 u(x_i) + O(h) \tag{4.20}$$

and for the $k = 2$ extension procedure:

$$\Delta^h u(x_i) = \Delta u(x_i) + \frac{h}{3}\sum_{d=1}^{n} \frac{1 - \kappa_d^2}{1 + \kappa_d}\partial_d^3 u(x_i) + O(h^2) \tag{4.21}$$

By Theorem 4.0.2, there exists a unique function $u^h \in \ell^\infty(\Omega^h)$ which satisfies

$$\Delta^h u^h(x_i) = \Delta u(x_i)\ \forall x_i \in \Omega^h \tag{4.22}$$

$$u^h(x_j) = u(x_j)\ \forall x_j \in \partial\Omega^h \tag{4.23}$$

Uniqueness follows from 4.0.2, and existence follows from the fact that this is a square system of linear equations which has a unique solution for any right hand side.

Then we have from 4.0.2:

$$\left|u^h - u\right|_{\ell^\infty(\Omega^h)} \leq C \left|\Delta^h(u^h - u)\right|_{\ell^\infty(\Omega^h)} \tag{4.24}$$

If we use the $k = 1$ extension procedure, we cannot bound the right hand side by a constant times h because of the $O(1)$ truncation error near the boundary. If we use the $k = 2$ procedure, the right hand side can be bounded by a constant times $h \min_\kappa \frac{1-\kappa^2}{1+\kappa}$. However, it is possible to improve this estimate using a comparison function argument, taking advantage of the fact that the truncation error is only large at a set of points which are $O(h)$ from the boundary. This is analogous to a distance weighted norm result for classical solutions to elliptic PDEs; see for example [6] Theorem 4.9 and 6.21. In [1], the authors show that we can decompose this error analysis into two pieces: the contribution from the interior and the contribution from points near the boundary. They employ a discrete Green's function in the proof, but their argument can be made with only a comparison function:

LEMMA 4.0.1. *Let $u^h \in \ell^\infty(\Omega^h)$ be such that $\Delta^h u^h(x_i) = 0$ at all interior points and $u^h = 0 \in \partial\Omega^h$. Then we have for the $k = 1, 2$ extension procedures:*

$$|u|_{\ell^\infty(\Omega^h)} \leq Ch^2 \left|\Delta^h u^h\right|_{\ell^\infty(\Omega^h)} \tag{4.25}$$

Proof: Let $M := \left|\Delta^h u^h\right|_{\ell^\infty(\Omega^h)}$. Define v^h to be equal to $\frac{M}{n}h^2$ at all lattice points in Ω^h, and zero on $\partial\Omega^h$. Then we have for the $k = 1$ procedure at a non interior point x_i:

$$\Delta^h v(x_i) = \frac{Mh^2}{nh^2} \sum_{d=1}^{n} \frac{-(1+\kappa_d)}{\kappa_d} + 1 \leq -M \tag{4.26}$$

and for the $k = 2$ procedure:

$$\Delta^h v(x_i) = \frac{Mh^2}{nh^2} \sum_{d=1}^{n} \frac{-2}{\kappa_d} + \frac{2}{\kappa_d + 1} \leq -M \tag{4.27}$$

and $\Delta^h v^h = 0$ at all interior points. Then we have that

$$\Delta^h u^h + v^h \leq M - M = 0 \tag{4.28}$$

$$\Delta^h u^h - v^h \geq -M + M = 0 \tag{4.29}$$

which by Theorem 4.0.1 implies

$$|u| \leq \frac{Mh^2}{n} \in \Omega^h \tag{4.30}$$

Thus we can split our error into two parts and recover global second order convergence:

THEOREM 4.0.3. *Let $\Omega \subset \mathbb{R}^n$ be an open connected set, and h be such that Ω^h satisfies our geometric assumptions. Let $u \in C^5(\Omega) \cap C^4(\overline{\Omega})$. Then the solution u^h to $\Delta^h u^h(x_i) = \Delta u(x_i) \in \Omega^h, u^h(x_i) = u(x_i) \in \partial\Omega^h$ satisfies*

$$|u - u^h|_{\Omega^h} \leq Ch^2 \tag{4.31}$$

where C depends only on diam(Ω), n, and bounds on higher derivatives of u.

Proof: Define B^h to be the set of non-interior points in Ω^h. Let $e^{h,I}$ and $e^{h,B}$ satisfy:

$$\Delta^h e^{h,I} = \Delta^h(u^h - u) \in \Omega^h \setminus B^h, \Delta^h e^{h,I} = 0 \in B^h \cup \partial\Omega^h \tag{4.32}$$

$$\Delta^h e^{h,B} = 0 \in \Omega^h \setminus B^h, \Delta^h e^{h,B} = \Delta^h(u^h - u) \in B^h \cup \partial\Omega^h \tag{4.33}$$

so that $u^h - u = e^h = e^{h,I} + e^{h,B}$. By Theorem 4.0.1, we have

$$\left|e^{h,I}\right|_{\Omega^h} \leq C_0 \left|\Delta^h(u^{h,I} - u)\right|_{\Omega^h \setminus B^h} \leq C_0 h^2 \sup_{\Omega^h} \left\{\frac{1}{12}\sum_{d=1}^{n} \left|\partial_d^4 u\right|\right\} \leq C_1 h^2 \tag{4.34}$$

where the truncation error estimate is from (4.19). By 4.0.1, we have

$$\left|e^{h,B}\right|_{\Omega^h} \leq C_2 h^2 \left|\Delta^h(u - u^h)\right|_{B^h} \tag{4.35}$$

For the $k = 1$ extension procedure we get from (4.20)

$$C_2 h^2 \left|\Delta^h(u - u^h)\right|_{B^h} \leq C_2 h^2 \sup_{\Omega} \{|\Delta u|\} \leq C_3 h^2 \tag{4.36}$$

and for the $k = 2$ extension procedure we get from (4.21)

$$C_2 h^2 \left|\Delta^h(u - u^h)\right|_{B^h} \leq C_2 h^2 \sup_{\Omega} \left\{\frac{h}{3}\sum_{d=1}^{n} \left|\partial_d^3 u\right|\right\} \leq C_4 h^3 \tag{4.37}$$

so that for the $k = 1$ extension we have:

$$|u^h - u|_{\Omega^h} = \left|e^{h,I} + e^{h,B}\right|_{\Omega^h} \leq \left|e^{h,I}\right|_{\Omega^h} + \left|e^{h,B}\right|_{\Omega^h} \leq C_1 h^2 + C_3 h^2 \leq Ch^2 \tag{4.38}$$

and for the $k = 2$ extension:

$$|u^h - u|_{\Omega^h} = \left|e^{h,I} + e^{h,B}\right|_{\Omega^h} \leq \left|e^{h,I}\right|_{\Omega^h} + \left|e^{h,B}\right|_{\Omega^h} \leq C_1 h^2 + C_4 h^3 \leq Ch^2 \tag{4.39}$$

which completes the proof. □

It is of note that for the $k = 1$ scheme, the stencil near the boundary is not even consistent, and yet we can achieve uniform second order accuracy.

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